

Rahul Ray

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MSc Data Science student at Leiden University, seeking a 2027 machine learning or data science internship or thesis placement.

Education

MSc Computer Science – Data Science *Leiden University, Netherlands* Sep 2025 – Aug 2027 (expected)
BTech Computer Science *SRM Institute of Science and Technology, Chennai* 2021 – 2025

Work Experience

Machine Learning Intern

Mar – Aug 2025 · Fox Solutions Pvt. Ltd., India

- Built a face anti-spoofing model (presentation attack detection) separating live faces from photo, video-replay and mask attacks, trained on public benchmarks plus an in-house dataset captured on the exact camera the system would run on in deployment.
- Selected MobileViT over a full ViT to fit the device compute budget, then took it from training through ONNX export to on-device deployment on an NVIDIA Jetson Nano, sustaining 30+ FPS real-time inference in a production-ready state.

Projects

Is it the Pruning, or the Ordering? – A Permutation-Controlled Re-evaluation of RAG

[writeup](#) [GitHub](#)

Python · PyTorch · Qwen2.5-3B · HotpotQA · LLMingua-2 · bootstrap CIs · pre-registration

- Pruning a RAG context also repositions the passages it keeps, so gains confound selection with position. Added two controls over 45,510 generations: five orderings per question, and a placebo dropping passages by position without reading them. Pruners beat the placebo by +0.2760 token-F1 (95% CI [0.222, 0.330]), refuting my own pre-registered hypothesis; only 35% of questions answer identically across orderings.

Attention Sinks and Quantization – An Audit of a Published Claim

[writeup](#) [GitHub](#)

Python · PyTorch · transformers · lm-eval-harness · FineWeb-Edu · LAMBADA · bootstrap CIs

- Audited whether architectural attention-sink mitigation buys robustness to low-bit activation quantization: 360 grid cells over five checkpoints differing by one boolean in `config.json`. The fix works under per-tensor 8-bit (55× less damage for 0.1% extra parameters) but is redundant under per-token scaling, which buys the same 32–46× for free; and the *lowest* sink-mass variant is the only one quantization destroys, traced to one layer-0 tensor.

Can LLMs Make Tabular ML Better?

[writeup](#) [GitHub](#)

Python · XGBoost · LLaMA 3.1 · sentence-transformers · Optuna · SHAP

- Tested LLM-generated semantic features as XGBoost input under nested CV (3 × 5-fold, leak-free PCA, 30 Optuna trials), on a second dataset picked to control for pretraining contamination; the hybrid never beat the baseline (HMEQ 0.865 vs 0.904 AUC, $p = 0.004$, $d = -0.84$), documented with Wilcoxon tests and SHAP rather than reframed.

AutoInsight – End-to-End AutoML Web App

[writeup](#) [GitHub](#)

Python · Streamlit · XGBoost · LightGBM · CatBoost · SHAP · Optuna · Groq API

- Turns a raw CSV into a ranked leaderboard, SHAP explanations and a plain-language summary with no code written: 7 models per task type in parallel, 40-trial Optuna TPE tuning, LLM fed aggregated metrics only so raw rows never leave the session. Live on Hugging Face Spaces.

Epoch-wise Double Descent Under Label Noise – ResNet-18 / CIFAR-10

[writeup](#) [GitHub](#)

Python · PyTorch · ResNet-18 · CIFAR-10

- Reproduced Nakkiran et al. (2019) over 4 noise rates × 3 seeds to 500 epochs; the corrupted-accuracy inflection tracks the test-loss peak (epoch 385 at $\eta = 20\%$, 473 at 30%). Label smoothing worsened final loss (0.87 vs 0.57), Mixup improved it (0.48 vs 0.57), both reported as found.

JEPA-IR – Benchmarking I-JEPA for Image Retrieval

[writeup](#) [GitHub](#)

Python · PyTorch · I-JEPA ViT-H/14 · CLIP · FAISS

- Benchmarked I-JEPA vs CLIP vs ResNet-50 as frozen extractors over 11.5k images and 110 queries (Oxford5k, Paris6k); I-JEPA reached 0.18 mAP against CLIP's 0.49, and near-zero query expansion gain (+0.002 vs +0.011) pinned the cause to embedding geometry, not tuning.

Text-JEPA – Porting the JEPA Objective from Vision to Language

[writeup](#) [GitHub](#)

Python · PyTorch · DistilBERT · WikiText-103 · EMA · SST-2 · STS-B · SVD geometry analysis

- Adapted I-JEPA (Assran et al. 2023) to language on DistilBERT, training a predictor to match EMA target-encoder representations at masked spans rather than predict masked tokens, 5000 steps on WikiText-103; anisotropy fell sharply (mean pairwise cosine 0.93 → 0.60) but downstream performance dropped (SST-2 81% → 75%, STS-B Pearson 0.48 → 0.16), a clean isotropy versus task-utility tradeoff.

Technical Skills

Core	Python, PyTorch, pandas, NumPy, scikit-learn, experiment design, statistical testing, nested CV, pre-registration
Tabular ML	XGBoost, LightGBM, CatBoost, SHAP, Optuna, feature engineering & selection, Wilcoxon, Cohen's d
Deep Learning	Vision Transformers, ResNet/CNNs, self-supervised learning, Mixup, label smoothing, generalisation theory
LLMs	Prompting & evaluation design, RAG, context pruning, quantization (per-tensor/per-token), attention analysis, lm-eval-harness
RL	Q-Learning, DQN, Policy Gradients, PPO, Actor-Critic, Gymnasium
NLP	sentence-transformers, TF-IDF, BM25, c-TF-IDF, topic modelling, NER, UMAP, HDBSCAN
IR	FAISS, dense retrieval, CLIP, multimodal retrieval, mAP, NDCG, Recall@k, query expansion, re-ranking
MLOps	Automated ML pipelines, edge deployment (ONNX, Jetson), model compression, heuristic & graph search, Hugging Face, Streamlit
Languages	English – C1 Professional Proficiency (<i>IELTS Certified</i>) Hindi – Native